

Chemistry

Unit 1

Practice Exam 1 Answers

Section one: Multiple-choice

Question 1

- D** In an atom, there are forces of attraction between the positively charged nucleus and the negatively charged electrons. Also, there are forces of repulsion between positively charged protons in the nucleus and between negatively charged electrons surrounding the nucleus.

Question 2

- D** Element X would have 4 valence electrons so needs 4 electrons to achieve a stable configuration. Element Y has 6 valence electrons and hence needs 2 electrons to achieve stability. Hence one X atom would bond with 2 Y atoms.

Question 3

- C** All atoms of the same element have the same number of protons in their nuclei but may have a different number of neutrons.

Question 4

- C** Elements in the same group of the periodic table have similar properties as they have the same number of valence electrons. Only N and P belong to the same group.

Question 5

- B** All atoms of copper have the same atomic number. However isotopes differ in the number of neutrons in the nucleus. The number of neutrons is the mass number minus the atomic number. So an atom of copper, atomic number 29, could have either 34 neutrons ($63 - 29$) or 36 neutrons ($65 - 29$).

Question 6

- C** By definition, the empirical formula shows the simplest mole ratio of each element in the compound. B is not correct as molecular compounds do not have ions.

Question 7

- A** A valency of -3 is obtained when an atom gains 3 electrons. An atom with the electron configuration of 2,5 would tend to gain 3 electrons to achieve a noble gas configuration.

Question 8

- A** An ionic solid consists of ions in fixed positions. When molten, these ions are free to move and hence carry a current.

Question 9

- B** The electrostatic forces of attraction between ions will determine the melting point of an ionic solid. These electrostatic forces of attraction are proportional to the charge on the ions. MgO is the only compound listed in which both ions are doubly charged ($2+$ and $2-$), and so these will exert the strongest attraction. So MgO will be expected to have the highest melting point.

Question 10

- D** The sodium ion is Na^+ and the sulfate is SO_4^{2-} .

Question 11

- A** Since ΔH is shown to be negative, the reaction must be exothermic. By definition, $\Delta H = E(\text{products}) - E(\text{reactants})$. As ΔH has a negative value, the energy of the bonds broken in the reactants must be greater than the energy of the bonds formed in the products.

Question 12

- C** If 0.25 mol has a mass of 19.5 g, then 1 mol will have a mass of 78.0 g. So the molar mass of the compound is 78.0 g mol^{-1} . The only compound with this molar mass is C_6H_6 .

Question 13

- D** Isotopes are atoms of the same element (hence the same number of protons) but differ in the number of neutrons in the nucleus.

End of section one

Section two: Short answer

(Note: In multi-mark questions each allocated mark is indicated as * and total marks are indicated in parenthesis.)

Question 14

- a** K_3PO_4 (1 mark)
 Al_2O_3 (1 mark)
- b** Calcium nitrate (1 mark)
Copper(II) carbonate (1 mark)

Question 15

- a i** Lattice of positively charged magnesium ions* surrounded by a 'sea' of valence electrons*. Lattice is held together by the electrostatic attraction between electrons and cations*. (3 marks)
- ii** The electrons are not localised/are free to move and so can conduct an electric current. (1 mark)
- b** (3 marks)

Solid ionic compounds cannot conduct electricity.	Though charged particles are present, they are held in fixed positions in the lattice so cannot move to carry a current*.
Ionic compounds are generally brittle.	If a strong force is applied to a crystal of the compound, the layers of ions will move relative to one another causing ions of like charge to be adjacent and hence repel. The crystal thus shatters*.
Ionic compounds have high melting temperatures.	A large amount of energy is needed to overcome the electrostatic attraction between oppositely charged ions*.

Question 16

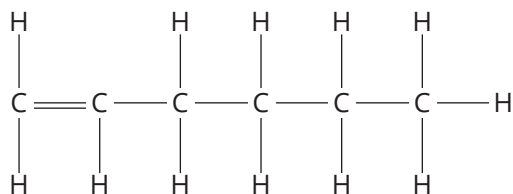
- a** The energy content of the reactants (HCl and NaOH) is greater than that of the products (NaCl and water)*. When HCl and NaOH react, the excess energy is released as heat and so the temperature of the reaction mixture rises*. (2 marks)
- b** When an element is heated, energy is absorbed and electrons are promoted to a higher energy level. When they return to a lower energy level, light is emitted*. The amount of energy absorbed and reemitted depends on the attraction between the electrons and the nuclear charge*. Different elements absorb, and hence emit, different energies as they differ in their nuclear charge and hence their electrostatic attraction between electrons and the nucleus*. (3 marks)

Question 17

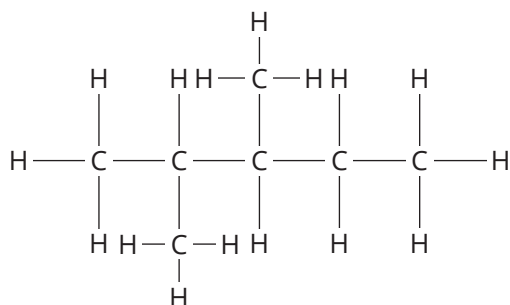
- a** pent-2-ene (1 mark)
heptane (1 mark)
methylbutane (1 mark)

b

(1 mark)



(1 mark)

**Question 18**

- a** $\text{CH}_3\text{CHCH}_2 + \text{Br}_2 \rightarrow \text{CH}_3\text{CHBrCH}_2\text{Br}$
(1 mark for showing correct structures of carbon compounds, 1 mark for correct equation)
- b** $2\text{C}_2\text{H}_6(\text{g}) + 7\text{O}_2(\text{g}) \rightarrow 4\text{CO}_2(\text{g}) + 6\text{H}_2\text{O}(\text{g})$
(1 mark for correct equation, 1 mark for correct balancing)
- c** $\text{C}_6\text{H}_6(\text{l}) + \text{Cl}_2(\text{g}) \rightarrow \text{C}_6\text{H}_5\text{Cl}(\text{l}) + \text{HCl}(\text{g})$
(1 mark for correct reactants, 1 mark for correct products)

Question 19

- a** $n(\text{Al}(\text{NO}_3)_3) = m/M = 30.5/213.01^* = 0.143^*$ (2 marks)
- b** $n(\text{N}) = 3 \times 0.143 = 0.430$ (1 mark)
- c** Total number of mol of atoms = $13 \times 0.143 = 1.86^*$
Total number of atoms = $1.86 \times 6.02 \times 10^{23} = 1.12 \times 10^{24}^*$ (2 marks)
- d** $\% \text{O}_2 = 9 \times 16/213.01 \times 100 = 67.6$ (1 mark)

Question 20

- a**
- | | | | | |
|-----------------------|---|--------------------|---|----------------------|
| $n(\text{C})$ | : | $n(\text{H})$ | : | $n(\text{O})$ |
| $\frac{54.54}{12.01}$ | : | $\frac{9.1}{1.01}$ | : | $\frac{36.36}{16.0}$ |
| $= 4.54$ | : | $= 9.1$ | : | $= 2.273^*$ |
| $= 2$ | : | $= 4$ | : | $= 1^*$ |
- So empirical formula is $\text{C}_2\text{H}_4\text{O}^*$. (3 marks)
- b** $n = m/M$ so $M = m/n = 30.8/0.350 = 88$ (1 mark)
- c** $\text{RFM}(\text{C}_2\text{H}_4\text{O}) = 44$
So molecular formula is $\text{C}_4\text{H}_8\text{O}_2$. (1 mark)

End of section two

Section three: Extended answer

Question 21

- a**
- i** 2,8,8,2 (1 mark)
 - ii** 2,8,8 (1 mark)
 - iii** 2,8 (1 mark)
 - iv** 1 calcium ion to 2 fluoride ions (1 mark)
 - v** Each calcium atom needs to lose 2 electrons to achieve stability but each fluorine atom only needs to gain one. Hence 2 fluoride atoms are needed for each calcium atom. (1 mark)
- b** The atomic radius of calcium is smaller than that of sodium*. Ca has a greater nuclear charge than Na but has the same number of occupied electron shells so electrons of Ca are drawn closer to the nucleus*. (2 marks)
- c** It is in an excited state because the third shell could accommodate the last electron and this is at a lower energy level. (1 mark)
- d**
- i** The sum of the number of protons and neutrons in a nucleus of an atom. (1 mark)
 - ii** The number of protons in a nucleus of an atom. (1 mark)
 - iii** Atoms with the same atomic number but different mass number, or atoms with the same number of protons but a different number of neutrons in the nucleus. (1 mark)
 - iv** A positively or negatively charged atom or group of atoms. (1 mark)
- e** O (1 mark)

Question 22

- a** Carbon dioxide is a covalent molecular compound*. It has weak forces holding molecules together and so has a low melting point*.
Silicon dioxide exists as a giant covalent lattice*. The strong covalent bonds throughout the lattice require high energy to break and so silicon dioxide has a high melting point*. (4 marks)
- b**
- i** They are all strong. (1 mark)
 - ii** Silicon dioxide is a lattice with covalent bonds between Si atoms and O atoms*.
Iron consists of cations surrounded by delocalised electrons. The bonding is metallic*.
Strontium fluoride consists of a lattice of positively charged Sr^{2+} ions and negatively charged F^- ions. The bonding is ionic*. (3 marks)
 - iii** Only iron conducts electricity in the solid state as only iron consists of charged particles that are free to move*. The electrons in the metallic lattice are delocalised and free to move* to carry a current. (2 marks)
 - iv** All bonding is electrostatic in nature. (1 mark)
- c** Any two ionic compounds such as NaCl, MgO, KBr, etc. (2 marks)

Question 23

- a**
- i** H_2O , C_3H_8 (1 mark)
 - ii** $\text{C}_6\text{H}_{12}\text{O}_6$, O_2 , C_5H_{10} (1 mark)
 - iii** O_2 , C_5H_{10} . Also accept CH_2 but no penalty if not included. (1 mark)
 - iv** C_{60} , $\text{C}(\text{graphite})$. Also accept O_2 and Si but no penalty if not included. (1 mark)
 - v** C_5H_{10} . Also accept $\text{C}_6\text{H}_{12}\text{O}_6$ but no penalty if not included. (1 mark)
 - vi** NaCl , CuSO_4 (1 mark)
 - vii** CH_2 , C_5H_{10} (1 mark)
 - viii** CuSO_4 (1 mark)
- b** Whilst stirring, add enough water to the mixture to dissolve the NaCl^* .
Filter the mixture to remove the insoluble $\text{C}(\text{graphite})^*$.
Evaporate the water from the filtrate to obtain the NaCl^* . (3 marks)

End of answers